ALG-01 - Teacher Diagnostic Key

Assimilation-book checkpoints

- Opening diagnostic x^4+3x under $x^2+x=1$: 2.
- H3 symmetric example: 104.
- H2 relation example: $x^5=55x-21$.
- H1 cubic symmetric example: $p^3+q^3=351$.
- H0 relation example under $t^2=2t+3$: $t^6-5t^4=82t+78$.

Recognition and first-line lab

1. $(103-97)(103+97)$.
2. Expand only enough to expose the coefficient: $(2x-3)(x+8)$.
3. $t=x+1/x$, retaining $x\neq 0$.
4. $a^2+b^2=(a+b)^2-2ab$.
5. Start with $x^2=4x-2$ and reduce after each multiplication.
6. Subtract the right-hand square and factor the difference of squares: $[(x+3)-(x-1)][(x+3)+(x-1)]=0$.
7. No. Zero-product structure is already visible.
8. $(u+v)^2=(u-v)^2+4uv$.
9. $x\neq -2$.
10. Multiply by t and use $t^2=-t-1$: $t^3=1$.
11. $p^3+q^3=(p+q)^3-3pq(p+q)$.
12. Retain/check $x=0$ before dividing by x.
13. $t=x^2$.
14. Expand.
15. Factor.
16. $(m-n)^2=(m+n)^2-4mn$ determines magnitude only; the sign depends on which variable is larger.
17. Begin from $y^2=5y-1$; multiply and reduce.
18. Substitute every candidate into the original equation before the implication-only step.

Practice bank answers/checkpoints

1. $(x-7)(x+7)$.
2. $4x^2-12x+9$.
3. 73.
4. $(52-48)(52+48)=400$.
5. Identity.
6. $8*2000=16000$.
7. $x^3=22x-15$, $x^4=95x-66$.
8. $t=y+1/y$, giving $t^2-7t+10=0$, with $y\neq 0$.
9. $8^3-3*12*8=224$.
10. $x=7,-2$.
11. $t^3=10t+3$, $t^4=33t+10$, $t^5=109t+33$, $t^6=360t+109$.
12. Move all terms to one side and factor: $(2x+3)^2-x^2=[(2x+3)-x][(2x+3)+x]=(x+3)(3x+3)=0$. Thus $x=-3,-1$. Subtracting equal expressions and factoring an identity are equivalence-preserving; no radical/domain doctrine is involved.
13. $(p+q)^2=(p-q)^2+4pq=25+56=81$.
14. Let $S_n=z^n+z^{-n}$. $S_1=3$, $S_2=7$, $S_3=18$, $S_4=47$.
15. Cancelling can lose $x=1$; preserve that branch before division.
16. $x^2=x+1$; successive reduction gives $x^8=21x+13$.
17. $a^3+b^3+c^3=3abc$.
18. $rs=(25-13)/2=6$; $r^4+s^4=(r^2+s^2)^2-2r^2s^2=169-72=97$.
19. $S_0=2,S_1=4,S_n=4S_{n-1}-S_{n-2}$; $S_5=724$.
20. Use difference of squares: $(x+6)^2-x^2=6(2x+6)=20$. Hence $12x+36=20$, so $x=-4/3$. Each step is reversible; no candidate-filter step is introduced.
21. Under $q^2=2q+1$: $q^3=5q+2$, $q^4=12q+5$, $q^5=29q+12$, $q^6=70q+29$, $q^7=169q+70$; subtract $13q^3=65q+26$, giving $104q+44$.
22. $(a-b)^2=36-20=16$, so $(a-b)^4=256$.
23. $w+1/w=2$ gives $(w-1)^2/w=0$, so $w=1$; target 2. (A recurrence also gives the same.)
24. $t=x^2+3x$; $t^2-7t+10=0$, so repeated block t is 2 or 5.
25. $x^2=1-x$, hence $x^4=2-3x$; target 2.
26. Any correct linear relation-reduction trace under $x^2=4x-1$; canonical polynomial modulo/remainder language is deferred to ALG-03.
27. With perimeter and area, $u+v$ and uv are known; symmetric powers such as u^2+v^2 can be reconstructed.
28. Accept a goal that treats the repeated pair as a structured object and chooses inputs that expose/cancel it; do not require functional-equation theory.
29. $(m+1)(n+1)=36$.
30. Retrieve ALG-01 target/representation selection; the inequality topic then owns the bound/equality/attainment machinery.

31-32. Evaluate whether the chosen representation is demonstrably cheaper for the stated target.

H0 control map for the learner-facing Independent Mixed Mastery Check

The H0 designation remains teacher/control metadata only and is not displayed in the learner-facing mastery title.

1. $(1007-993)(1007+993)=14*2000=28000$.
2. $13^3-3*36*13=793$.
3. $x^3=9x+10$, $x^4=28x+45$, $x^5=101x+140$.
4. $p^2+q^2=(p-q)^2+2pq=16+42=58$.
5. Subtract the squares and factor: $[(3x+4)-(x+2)][(3x+4)+(x+2)]=(2x+2)(4x+6)=0$. Thus $x=-1,-3/2$. The transformation is an identity-equivalent rewrite, so it creates no extra candidates.
6. $S_3=5^3-3*5=110$.
7. From $u^2+u+1=0$, $u^3=1$; 2026 mod 3=1, so $u^{2026}=u$.
8. Example $t=x^2+3x$; it is useful when rewriting in t reduces repetition/degree and recoverable restrictions remain visible.
9. Division assumes $x\neq 0$ and therefore discards a valid original solution.

10. Expand strategically; the target is coefficient information. Full expansion is $x^3+6x^2+11x+6$, coefficient 6.
11. Factor: $(x-3)(x-7)=0$; zeros 3,7.
12. Only $(m-n)^2=81-56=25$ is determined; without an order convention $m-n$ can be 5 or -5.
13. $t^3=15t-4$, $t^4=56t-15$; target $41t-15$.
14. Exclude $x=3$. Multiply through: $x+1=2x-6$, so $x=7$; original check passes.
15. Cheaper first line: $r^2+s^2=(r+s)^2-2rs$; result $64-30=34$.
16. $(a+1)(b+1)=49$.

Diagnostic codes

- ALG01-R1: manipulates before reading target.
- ALG01-R2: factor/expand direction confusion.
- ALG01-R3: substitution does not reduce complexity.
- ALG01-R4: symmetric target solved through unnecessary individual values.
- ALG01-R5: relation not used as a rewriting rule.
- ALG01-R6: equivalence/condition check missing.
- ALG01-R7: downstream canon imported (Vieta/discriminant/remainder/AM-GM/radical doctrine).

Remediation routing

- R1/R2 -> First-Step router + contrast items 10/11.
- R3 -> substitution tests in the Assimilation Book.
- R4 -> symmetric reconstruction contrast.
- R5 -> relation-rewrite examples, then mastery #13.
- R6 -> equivalence section, then mastery #5/#14.
- R7 -> route learner to the canonical downstream topic rather than reteaching it here.